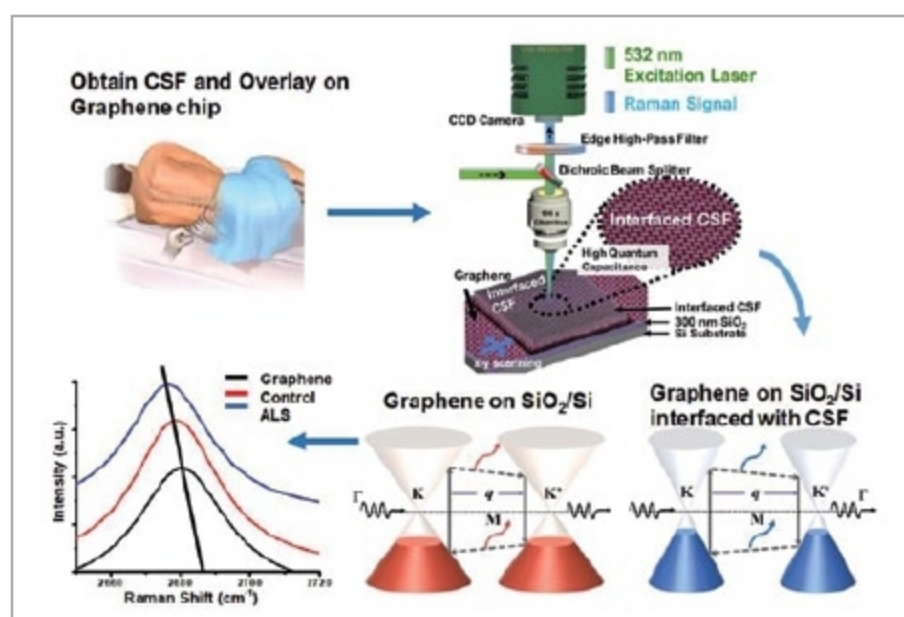


Asegun Henry, the Robert N. Noyce Career Development associate professor in department of mechanical engineering, says, "The reason that this technology is interesting is, once you do this process of focussing light to get heat, you can store heat much more cheaply than you can store electricity."

Using graphene to detect ALS, other neurodegenerative diseases

According to new research from University of Illinois at Chicago, supermaterial graphene could one day be used to test for amyotrophic lateral sclerosis (ALS)—a progressive, neurodegenerative disease that is diagnosed mostly by ruling out other disorders.



How graphene can be used to detect ALS biomarkers from cerebrospinal fluid (Credit: University of Illinois at Chicago)

When cerebrospinal fluid from patients with ALS was added to graphene, it produced a distinct and different change in the vibrational characteristics of the graphene compared to when fluid from a patient with multiple sclerosis (MS) was added, or when fluid from a patient without any neurodegenerative disease was added to graphene. These distinct changes accurately predicted what kind of patient the fluid came from—one with ALS, MS or no neurodegenerative disease.

Graphene is a single-atom-thick material made up of carbon. Each carbon atom is bound to its neighbouring carbon atoms by chemical bonds. Elasticity of these bonds produces resonant vibrations—also known as phonons—which can be very accurately measured. When a molecule interacts with graphene, it changes these resonant vibrations in a very specific and quantifiable way.

New antenna evaluation method that could help boost 5G network capacity

Researchers at National Institute of Standards and Technology (NIST), USA, have developed a method for evaluating and selecting optimal antenna designs for future 5G cellphones, other wireless devices and base stations. The new NIST

method could boost 5G wireless network capacity and reduce costs.

Antenna beam width affects wireless system design and performance. NIST's new measurement-based method allows system designers and engineers to evaluate the most appropriate antenna beam widths for real environments.

NIST engineer Kate Remley says, "Our new method could reduce costs by enabling greater success with initial network design, eliminating much of the trial and error that is now required. It would also foster the use of new base stations that transmit to several users either simultaneously or in rapid succession without one antenna beam interfering with another. This, in turn, would increase network capacity and reduce costs with higher reliability."

This is the first detailed measurement-based study of how antenna beam width and orientation interact with the environment to affect millimetre-wave signal transmission.



NIST researcher Jelena Senic drives a robot used to measure the performance of different antenna beam patterns. The mobile platform enables researchers to position a wireless channel sounder that includes (top to bottom) an array of 16 receive antennas, receiver, timing circuitry, signal digitiser and battery for untethered field operations (Credit: NIST)

In the technique, NIST measurements covering a broad range of antenna beam angles are converted into an omnidirectional antenna pattern covering all angles equally. The omnidirectional pattern can then be segmented into even narrower beam widths. Users can evaluate and model how antenna beam characteristics are expected to perform in specific types of wireless channels.

To develop the method, the team collected experimental data in a hallway and lobby of a NIST research building, using